Joy D'Souza

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Education

Georgia Institute of Technology, Atlanta, GA August 2024 – May 2028

Bachelor of Science in Mechanical Engineering; GPA: 4.0/4.0

Coursework: System Dynamics, Design and Manufacture, Statics, Deformable Bodies, Thermodynamics, Circuits Lab

Work Experience

SharkNinja, Product Test Engineering Intern, *Needham, MA* May 2026 – July 2026

- Developed a custom annotation application to quantify hair curl performance, establishing 6 objective KPIs from 100+ annotated hair tress images; validated measurement repeatability across technicians using Gage R&R and standardized application use through work instructions.
- Designed and assembled a standardized curl-retention test rig for 15 hair tresses, establishing camera alignment, lighting, and styling procedures for repeatable testing.
- Analyzed 50+ test results across 7 performance metrics, using one-way ANOVA in Minitab to evaluate statistical parity among build iterations and competitor products.

SharkNinja, Quality Engineering Intern, *Needham, MA* May 2025 – August 2025

- Performed hands-on teardowns and root cause analysis (RCA) on 50+ returned units to test and identify failures with PCBAs, motors, and other internal mechanisms; documented findings, testing methods, and corrective actions.
- Conducted global cross-functional collaboration with engineering, manufacturing, marketing, product development, and customer experience teams to contribute to corrective actions with Six Sigma methods, including a rework to enhance RF signal strength between the remote and receiver on a key home environment product.

Projects

Force Sensor for 3D-Printed Prosthetic, Medical Robotics Club September 2025 – Present

- Creating Force Sensing Resistor (FSR) calibration standard operating procedure to standardize sensor variability for force sensor integration into low-cost 3D-printed prosthetic for trans-radial amputees.
- Calibrating FSRs by applying known weights, collecting analog voltage data via Arduino, and analytically defining touch thresholds (soft, medium, hard) as prosthetic feedback for the user.
- Designing and fabricating a 3D-printed FSR integration housing with silicone for comparative testing of force-transmission accuracy within a finger.

Sustainable Menstrual Pad, EWB-Wish For WASH October 2024 – Present

- Developing a sustainable menstrual pad with a 14-member bioplastics team to match absorbency, comfort, and cost of industry standards (benchmarked against Always pads).
- Identified Himalayan nettle and wood pulp as top-performing core materials based on mass gain by testing 9 candidate materials under ASTM- and ISO-based absorbency protocols using water and gelatin blood simulant.
- Prototyping a clasping mechanism to replace traditional adhesives in pads and improve biodegradability.

Mechanical Gripper with Machine Vision, Undergraduate Research January 2026 – May 2026

- Worked in a 2-person team to integrate a Raspberry Pi 5, machine vision system, and custom mechanical gripper with a UR5 robotic arm for industrial pick-and-place applications.
- Designed 3D-printed claw modification for existing gripper mechanism to improve object grasping performance.

Autonomous Competition Robot, ME2110 September 2025 – November 2025

- Collaborated with 3 team members to design and fabricate a fully-autonomous robot in 8-weeks under a $120 budget using provided mechatronics components and Arduino Uno for competition against 72 teams.
- Leveraged Onshape, Inkscape, and Bambu Studio for component fabrication on Trotec Laser machines and BambuLab P1S printers.
- Documented concept generation and evaluation, quality function deployment, and design decisions through technical reports and presentations using Morphological Charts, Engineering Specification Sheets, and House of Quality.

Skills

Technical: Root Cause Analysis, 8D Problem Solving, FMEA
Design: SolidWorks, Onshape, Bambu Studio, Inkscape
Equipment: 3D Printing, Hand Tools, Laser Cutting, Soldering, Instron, Mark-10, Oscilloscope, Function Generator
Computer Literacy: MATLAB, Java, Python, HTML, Jira, Zendesk, Microsoft Office, Minitab